

Ecosystem modeling: Individual-based model

I. Particle tracking algorithm

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May, 2026

Recap

How to develop a (numerical) ecosystem model?

1. Select model variables (e.g., phytoplankton, P).
2. Design governing equations for the variables:

$$\frac{dP}{dt} = \sum S_o - \sum S_i = UP - \sigma'P \quad (1)$$

3. Solve it (numerically).

In NPZD model, variable P represents “bulk” phytoplankton (as a functional group), corresponding to the population.

So, the approach we discussed before is referred to as the **population dynamics modeling**, limited to resolve each individual's behavior. We can set up each equation for each individual, so-called **individual-based modeling**.

Individual-based model

One of the simplest (but powerful) individual-based model is particle tracking algorithm¹:

(Simple) particle tracking algorithm

$$\frac{dX_i}{dt} = U_i \quad (2a)$$

$$\frac{dY_i}{dt} = V_i \quad (2b)$$

where

X_i and Y_i : x - and y -position of i -th individual (particle)

U_i and V_i : each velocity component at the position (X_i, Y_i) .

It is worth noting that velocity field $u(x, y)$ and $v(x, y)$ are required, so $U_i = u(X_i, Y_i)$ and $V_i = v(X_i, Y_i)$.

¹Note that there are no biological terms yet.

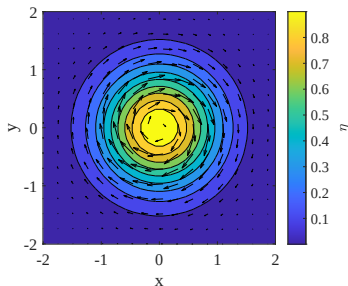
Lab 1

Consider two-dimensional velocity field governed by geostrophic balance:

$$\eta = \eta_0 e^{-(x^2+y^2)/L^2} \quad (3a)$$

$$u = -\frac{g}{f_0} \frac{\partial \eta}{\partial y} \quad (3b)$$

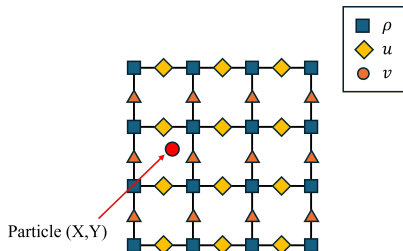
$$v = \frac{g}{f_0} \frac{\partial \eta}{\partial x} \quad (3c)$$



Note that this is just system of ordinary differential equations (ODEs), so we can still use ODE solver to solve it!

A more practical model

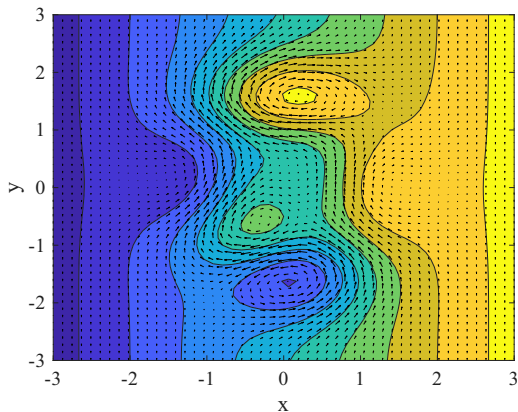
We never know the exact velocity fields (as a function). In most cases, we only have a limited number of sampled estimates (e.g., observations and discretized velocity fields from hydrodynamics model).



U_i and V_i need to be **interpolated** from the sampled u and v fields, respectively. Make sure to change the coordinate if the velocity field data is given in spherical coordinates.

Lab 2

Develop your own particle tracking code and launch particles on the velocity field data.



A more practical model

So, particle tracking algorithm has two error sources: one from numerical discretization (truncation error) and the other from interpolation.

There are many open-source particle tracking models (softwares), well summarized by Van Sebille et al. (2018).

- ▶ Döös, Jönsson, and Kjellsson (2017) developed analytical time integration method for linear interpolation, built in TRACKMASS (available at <https://www.tracmass.org/>)
- ▶ Delandmeter and Van Sebille (2019) highlighted that choice of interpolation method, depending on grid system (A- or C-grid), is important². This is considered in PARCELS (available at <https://parcels-code.org/>).

²See also https://www.youtube.com/watch?v=QKSbbi_67Ro&t=1712s

Further information

Note that the governing equation of the particle tracking algorithm (2) is equivalent to the advection equation³:

$$\frac{dX}{dt} = U, \quad \frac{dY}{dt} = V \quad (4)$$

(Individual-based model)



$$\frac{\partial C}{\partial t} + u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} = 0. \quad (5)$$

(Population dynamics model)

³https://jang-geun.github.io/vis_geo_advection2.gif

References I

-  Delandmeter, Philippe and Erik Van Sebille (2019). “The Parcels v2. 0 Lagrangian framework: new field interpolation schemes”. In: *Geoscientific Model Development* 12.8, pp. 3571–3584.
-  Döös, Kristofer, Bror Jönsson, and Joakim Kjellsson (2017). “Evaluation of oceanic and atmospheric trajectory schemes in the TRACMASS trajectory model v6. 0”. In: *Geoscientific Model Development* 10.4, pp. 1733–1749.
-  Van Sebille, Erik et al. (2018). “Lagrangian ocean analysis: Fundamentals and practices”. In: *Ocean modelling* 121, pp. 49–75.